

Nicolas Adler

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EDUCATION:

University of Miami | 2023-2025

Coral Gables, FL

Master of Science in Music Engineering Technology

- **GPA:** 3.75
- **Honors:** Dean's List, Member of the Acoustical Society of America
- **Relevant Coursework:** Digital Signal Processing 1,2 and 3 (currently), Acoustics, Transducers Audio Electronics, Research Techniques in Music Technology, Machine Learning with PyTorch and Scikit-Learn, and Music and Artificial Intelligence

University of Miami | 2018-2022

Coral Gables, FL

Bachelor of Arts in Instrumental Music Performance

- **GPA:** 3.5
- **Honors:** 2021 Concerto Competition Winner, and performance with Gerard Schwarz.

PROJECTS:

STI Calculator App:

- Developed a MATLAB app that calculates Speech Transmission Index (STI) by estimating reverberation time (RT60) from user inputs such as room volume and surface absorption coefficients. Tested in five rooms using an NTI meter, with a margin of error of less than 0.05 STI points.

Acoustic Treatment:

- Conducted physical and frequency measurements of a room, calculated absorption coefficients and RT60 times using Room EQ Wizard and a few of Sabine's equations to develop and implement an acoustic treatment plan. Despite the poor dimensions of the room that failed to meet Bolt's Area, the final model predicted a reduction of over 50% in the room's longest reverb time, creating a flatter frequency response and a room that suited its musical purpose.

DSP MATLAB Scripts and JUCE Plugins:

- Applied concepts from 'Digital Audio Theory' by Christopher Bennett and knowledge of developing MATLAB function and classes to complete chapter projects covering a range of topics including aliasing, delay effects, tremolo, vibrato, signal windowing, and various DSP applications.

Spanish Dialect ML model:

- Wrote a Python program that trained a CNN to classify various Spanish dialects: Argentinian, Chilean, Dominican, Mexican, and Spanish.

STI Master's Thesis:

- Led a research project improving Speech Transmission Index (STI) prediction for concert halls; collected calibrated impulse responses (REW), wrote MATLAB scripts for RT60, C50, D50, and STI analysis, and developed MLR, GPR, and MLP models. Reduced prediction error by **33%** compared to traditional methods.

WORK EXPERIENCE:

Motorola Solutions, Inc. Audio Software Engineer Co-Op

Fort Lauderdale, FL | Jan 2025 – Present

- Conducted acoustic testing and signal analysis for voice communication devices (SNRI, linearity, speech intelligibility).
- Programmed MATLAB scripts for speech quality analysis, automated test pipelines, and acoustic parameter extraction.
- Acoustic Utilized tools like OpenGrok and Bitbucket to navigate and make edits to the existing C++ codebase.
- Supported signal path evaluation for walkie-talkies across varied environmental and hardware configurations.

ACENTECH Audio Visual Technology Intern

- *Worked full time at the Cambridge office assisting in making edits to architectural documents specifically about all things AV and technology using software such as Revit, AutoCAD, Sketchup, Soundvision and Excel.*
- *Created a MATLAB app, followed by its subsequential classes, functions, and scripts to estimate speech transmission values, analyzed data for insights or trends, and footfall data.*

SKILLS:

- Programming: MATLAB, C++, Java, JavaScript, Python, SSH, OpenGrok, Bitbucket
- Audio DSP: Filtering, RT60 Measurements, STI Analysis, Signal Analysis
- Software Tools: OpenGrok, Bitbucket, JIRA, Revit, AutoCAD, Sound-Vision, Room EQ Wizard (REW)
- Other: Bilingual (English and Spanish)